

Multimodal, Storage-Aware Food Freshness & Shelf-Life Prediction

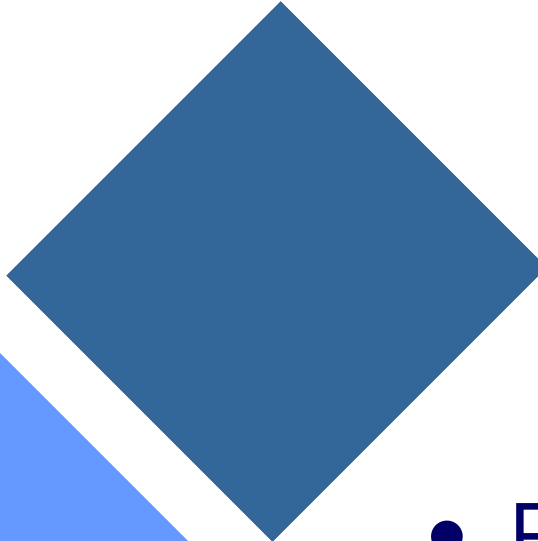
Submitted by:

- 23BCE7099 Meghana Sri Lakshmi Iswarya
- 23BCE8928 Lingineni Yasaswini
- 23BCE8955 VenkataSai Shreya Maturi
- 23BCE9513 Gaddam Madhuri

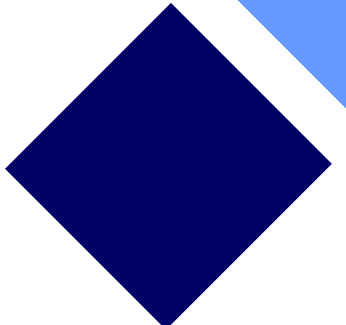
Guided by:

Dr. Balusa Bhanu Chander
Associate Professor Grade 1
SCOPE Dept





Introduction



- Food spoilage is a major contributor to food waste, especially in fruits and vegetables.
- Spoilage depends not only on the appearance of food but also on storage conditions such as temperature and humidity.
- Existing AI systems mainly analyze food images to identify the current freshness or spoilage stage.
- Image-only systems do not consider the environment in which food is stored.
- Our project combines visual and environmental information to provide more practical freshness and shelf-life predictions.

Problem Statement

Existing image-based food freshness systems have several limitations:

- They rely mainly on a single food image.
- They do not consider temperature, humidity, or storage conditions.
- They mainly identify the current spoilage state rather than future spoilage.
- They provide limited information about remaining shelf life.
- They usually do not provide actionable recommendations to users.

Our Goal:

To develop an AI system that predicts current freshness and future spoilage by considering both the food's appearance and its storage environment.

Literature Review

Approach	Method	Limitation
Image-based freshness detection	CNN-based models	Mainly focuses on visual appearance
Fruit/vegetable classification	ResNet, VGG, MobileNet	Limited to classification
Spoilage & shelf-life prediction	CNN + Transformer	Mainly uses image information
Sensor-based freshness detection	Temperature, humidity, gas sensors	Requires additional hardware
Explainable AI	LIME, Grad-CAM	Explains predictions but provides limited forecasting

Observation

Existing approaches provide useful freshness detection, but there is a need for a system that combines visual information + environmental conditions + future spoilage prediction.

Identified Gaps

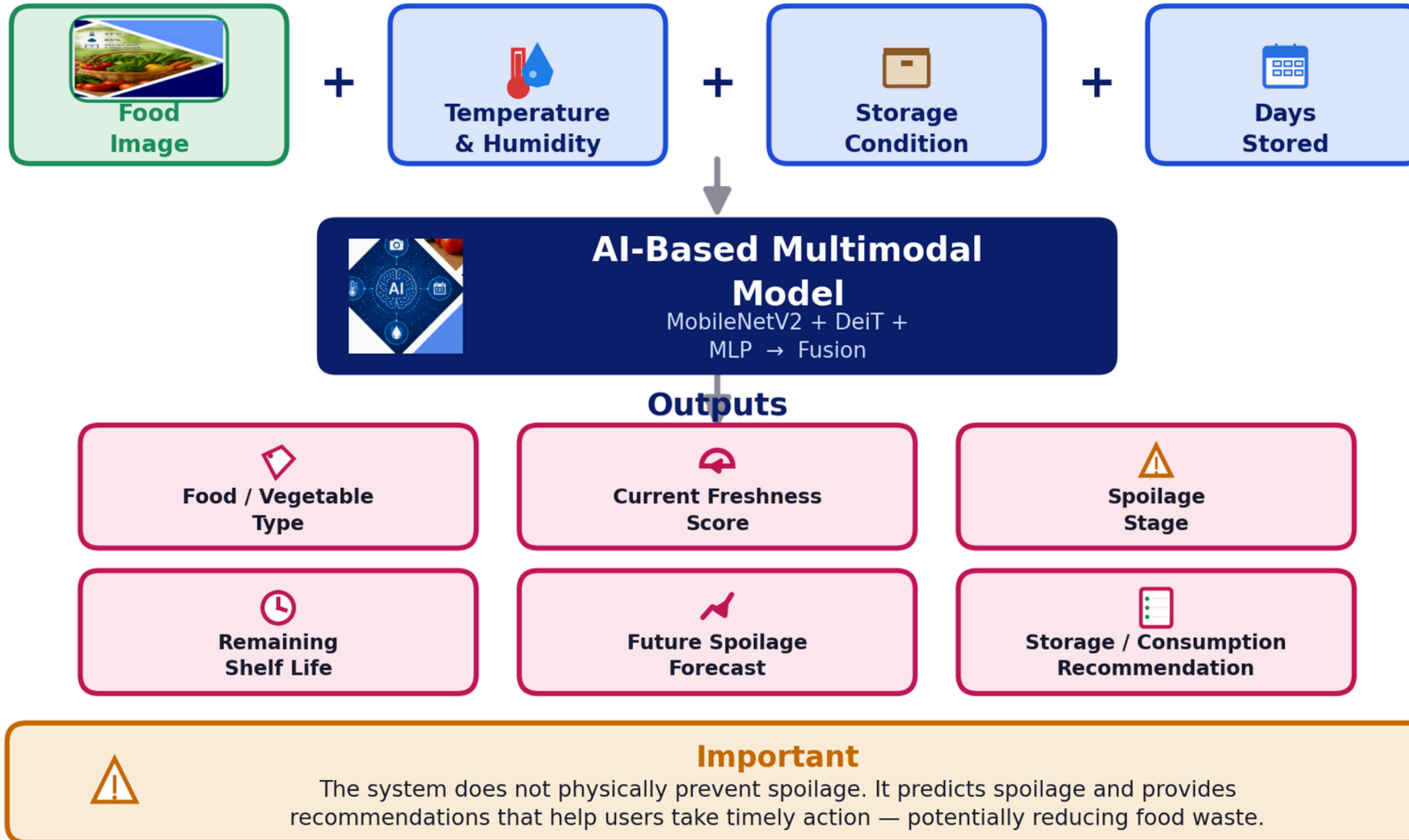
- No environmental context: Image-based models do not consider temperature and humidity.
- Static prediction: Most systems identify the current condition but do not forecast future spoilage.
- Limited freshness information: Coarse labels such as fresh/rotten provide limited detail.
- Limited decision support: Existing systems provide predictions but little practical guidance.

Research Opportunity

Develop a multimodal, storage-aware AI system that combines food images with environmental and storage information.

Proposed System — Core Concept

How the system predicts food freshness and spoilage

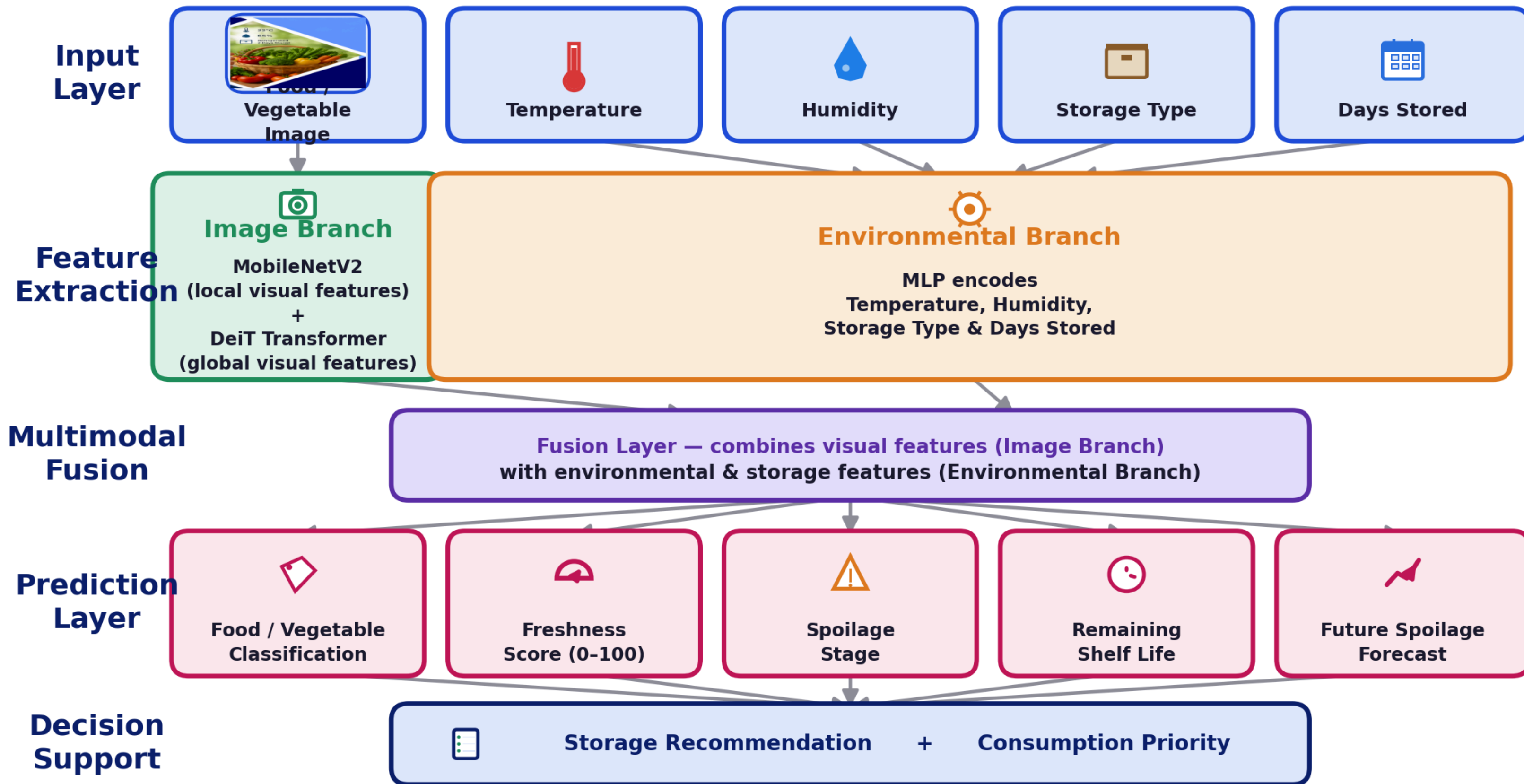


Objectives

1. Develop a multimodal AI model combining image and environmental information.
2. Predict a continuous freshness score for fruits and vegetables.
3. Identify the current spoilage stage.
4. Estimate the remaining shelf life under different storage conditions.
5. Forecast the future spoilage trend for upcoming days.
6. Provide storage and consumption recommendations based on the prediction.

Proposed System Architecture

Multimodal, Storage-Aware Food Freshness & Shelf-Life Prediction



Methodology

Step 1 – Data Collection

- Collect food images along with temperature, humidity, storage condition and storage duration.

Step 2 – Data Preprocessing

- Resize and normalize images, perform augmentation, and preprocess environmental data.

Step 3 – Feature Extraction

- Extract visual features using MobileNetV2 and DeiT and environmental features using an MLP.

Step 4 – Multimodal Fusion

- Combine visual and environmental features into a unified representation.

Step 5 – Model Training

- Train the model for freshness prediction, spoilage classification, shelf-life estimation and forecasting.

Step 6 – Evaluation & Deployment

- Evaluate model performance using appropriate metrics and develop a user interface for predictions and recommendations.

Novelty / Existing vs Proposed

Feature	Existing System	Proposed System
Input	Food Image	Image + Environment
Temperature / Humidity	✗	✓
Current Freshness	✓	✓
Spoilage Classification	✓	✓
Freshness Score	Limited	0–100 Score
Shelf-Life Estimation	Limited	Storage-Aware
Future Spoilage	✗	✓
Recommendations	Limited	✓
Decision Support	✗	✓

Combining visual features with storage conditions to make freshness and shelf-life prediction more context-aware and actionable.



Tools & Technologies

Category	Technologies
 Programming & ML	Python • PyTorch • TensorFlow • Scikit-learn
 Deep Learning	MobileNetV2 • DeiT • Multi-Task Learning
 Data Processing	NumPy • Pandas • OpenCV
 Environmental Data	Temperature Sensor • Humidity Sensor
 Explainable AI	LIME • Grad-CAM
 Deployment	Flask • HTML • CSS • JavaScript • SQLite

Expected Outcomes

- Accurate prediction of current food freshness.
- Identification of the spoilage stage.
- Estimation of remaining shelf life based on storage conditions.
- Forecast of the future spoilage trend.
- Personalized storage and consumption recommendations.



Conclusion

The proposed system aims to move beyond image-only freshness detection by combining food appearance with environmental and storage information. By predicting spoilage earlier and providing actionable recommendations, the system can help users make better food-storage and consumption decisions and potentially reduce avoidable food waste.



References

1. Base paper: Multimodal Deep Learning for Vegetable Classification, Spoilage Detection, and Shelf-Life Prediction using CNN-Transformer Fusion.
2. Sandler, M. et al., MobileNetV2: Inverted Residuals and Linear Bottlenecks, CVPR, 2018.
3. Touvron, H. et al., Training Data-Efficient Image Transformers & Distillation Through Attention, ICML, 2021.
4. Ribeiro, M. T., Singh, S., & Guestrin, C., Why Should I Trust You? Explaining the Predictions of Any Classifier, KDD, 2016.
5. Relevant research papers on food freshness detection, spoilage prediction and sensor-based monitoring.



THANK YOU